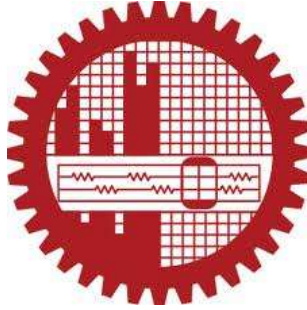


Bangladesh University of Engineering and Technology (BUET)
Department of Industrial and Production Engineering (IPE)



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**Project Report 2: Deep Reinforcement Learning for Inventory
Optimization under Non-Stationary Demand: Evidence and
Bangladesh Adaptation**

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1. Research Gap

The selected paper aims to cover a very critical gap existing in the inventory management literature. Most of the earlier inventory models, including a few early deep reinforcement learning (DRL) papers, have relied on the assumption of a stationary demand, i.e., the demand tends to stay the same over time. This assumption, however, does not hold true in most actual business environments. As the selected paper clearly states, the assumption of a stationary demand does not hold true most of the time, highlighting a critical gap between the actual environment and the earlier research models.

The next gap is that although DRL has been successfully applied to some inventory control problems, most of the previous research was based on simpler problem structures. The previous research demonstrated that DRL can achieve good performance in classical inventory systems, but it was still based on a stationary demand environment or a simple problem structure. Therefore, there was still a lack of research on how DRL can be used to deal with non-stationary uncertain demand environments, which means that forecasts vary over time. This paper aims to fill this gap by applying DRL to a more realistic forecasting environment.

The paper has also recognized a gap in the methodology used to solve these problems, which are commonly solved through a specific method. In a rolling-horizon non-stationary inventory system, a commonly used method is the repeated solution of a finite optimization model at each time period. This method, although acceptable, is not practical when the decisions have to be made quickly, as the process can become computationally burdensome, especially if the decisions are required to be made repeatedly. In a real-world context, it is always better to have a policy that can make quick decisions based on the changing forecast, rather than repeatedly re-optimizing the model.

Another gap that has not been addressed by researchers is the evolution of the forecast. Most businesses do not rely on a single forecast of demand, but the forecast is continuously revised based on new information received. Most models have neglected the evolution of the forecast, failing to integrate the new forecast information into the inventory decisions. This paper has contributed by incorporating a rolling forecast horizon, as well as the forecast evolution through the Martingale Model of Forecast Evolution (MMFE). This has made the problem setting closer to the real-world operations of the supply chain, a gap that was not addressed by the earlier researchers.

At the same time, the paper identifies that there is a gap in the literature regarding the ability of the DRL model to generalize. A good DRL model should be able to perform optimally not just for the scenarios that were experienced during training, but also for the scenarios that were not experienced during training. This is important because in the real world, supply chain systems always have new patterns and uncertain environments. The authors aim to prove that if the DRL agents are properly trained, they can achieve near-optimal performance for scenarios that were not experienced during training. This is a significant contribution because the previous literature had not adequately shown this for non-stationary inventory systems.

In brief, it can be concluded that the main gap in the paper is that the previous studies on inventory optimization failed to properly address a solution using DRL to tackle a single-item inventory problem under non-stationary uncertain demands with a rolling forecast horizon. In other words, the previous studies adopted stationary demands, used a less complex inventory system, or repeated solving a dynamic programming problem. However, the paper attempts to address this gap and propose a solution using a DRL-based approach.

2. Research Objectives and Questions

2.1 Research Objectives

The main objective of the selected paper is to propose a more practical and adaptive approach to decision-making in the context of inventory. In other words, it is to propose a more practical and adaptive approach to decision-making in a context where uncertainty is a major factor. The main focus of the selected paper is to propose a solution to a specific problem in a single-item lot sizing problem that is more relevant to a real-world environment due to various factors such as a fixed ordering cost, lead time, and backorders as well as lost sales. However, instead of a stationary environment with a fixed and stable demand, a non-stationary environment with a rolling forecast period is considered, which is more realistic. The main purpose of the selected paper is to assess if a deep reinforcement learning approach can be used to propose a good policy.

The second important goal that the paper pursues is to generalize previous research on the application of DRL in inventory control, which was limited to stationary demand. This is particularly obvious from the introduction, in which the authors indicate that previous research on the application of DRL in inventory control was mostly limited to classical systems with stationary demand, while in reality, many systems are characterized by non-stationary demand. This is an important goal, as it allows the application of DRL to move from an idealized to a more realistic environment.

Another objective of the research is to evaluate if a DRL agent can make nearly optimal decisions in real time without having to solve the optimization problem repeatedly. In a traditional rolling horizon inventory model, firms have to solve a finite horizon problem repeatedly. This is computationally intensive. The research aims to prove that if a DRL agent is well-trained on randomly generated environments, it can directly provide good quality decisions on previously unseen demand forecasts. This means that the research is not only concerned with theoretical performance but also with computational efficiency.

Another aim of the study is to test whether a DRL-based agent can make near-optimal decisions in real time without repeatedly solving the optimization problem. In the traditional rolling-horizon inventory problems, firms have to repeatedly solve the optimization problem at each period, which can be time-consuming. The aim of the paper is to demonstrate that after being properly trained in randomly generated environments, a DRL-based agent can directly provide

high-quality ordering decisions for unseen demand forecasts. This suggests that the paper is not only concerned with the theoretical performance of the model but also with the efficiency of the model's computation.

A further objective is the assessment of the applicability of DRL in various operating environments that consider both backorders and lost sales. This enhances the paper's relevance because various industries have unique service environments. In some industries, backorders can be employed as a means of delay in service fulfillment, while in other industries, lost sales occur if the product or service is unavailable for fulfillment.

2.2 Research Questions

Based on the paper, the main research questions can be written as follows:

RQ1: Can deep reinforcement learning be effectively applied to single-item inventory optimization problems under non-stationary uncertain demand?

RQ2: Can a DRL agent learn near-optimal replenishment policies when demand forecasts evolve under a rolling-horizon setting?

RQ3: Can trained DRL agents generate real-time ordering decisions for previously unseen forecast instances without repeatedly re-solving the optimization model?

RQ4: How does the performance of DRL-based inventory policies compare with dynamic programming-based heuristics in terms of solution quality?

RQ5: Does the proposed DRL framework remain effective under different inventory environments such as backorders and lost sales?

RQ6: Can the proposed methodology support practical inventory control problems with realistic assumptions such as fixed ordering costs, lead time, and changing forecasts?

Overall, it can be seen that the objectives and questions of the paper are closely related. The paper is intended to explore if DRL has a potential to become a solution tool for inventory systems with uncertain, non-stationary, and forecast-driven demand. The paper is not intended to merely prove a concept, but to demonstrate that DRL can achieve good results with a much more realistic supply chain environment. Therefore, it is clear that all objectives of methodology development, benchmark comparison, and validation are closely related.

3. Methodology

3.1 Overview of the Methodology

The paper introduces a methodology to tackle the non-stationary Single-Item Stochastic Lot-Sizing Problem (SISLSP) using a technique called deep reinforcement learning. The problem environment is considered to be non-stationary and is embedded with various characteristics of a practical inventory system, such as a fixed ordering cost, lead time, and backordering and lost sales. However, unlike other stationary SLPs, the problem is considered under a rolling forecast environment, in which information regarding future demands over a finite number of periods is available.

The methodology of the paper is explained in a logical order. First, the problem is mathematically formulated. Second, an approximate dynamic programming is designed to be used as a benchmark. Third, the problem is reformulated as a reinforcement learning problem. After that, the DRL agent is learned using the Proximal Policy Optimization (PPO) algorithm. At the end, the learned policy is evaluated through various experimental scenarios such as backorders, lost sales, and updating forecasts using the MMFE procedure.

3.2 Problem Definition

The study is based on a single-item inventory control problem with non-stationary uncertain demand. In every period, a decision has to be made on whether to order or not, and how much to order, based on the available inventory, orders received, future demand forecasts, and various costs associated with inventory management.

The main assumptions of the problem include:

- fixed ordering cost,
- holding cost,
- shortage or stock-out cost,
- lead time,
- rolling demand forecast horizon.

This problem setting is important because it reflects a more realistic inventory system than many classical models, where demand is often assumed to remain stationary over time.

3.3 Mathematical Formulation of the Inventory System

The methodology starts with the mathematical formulation of the system. At each decision period, the system is aware of the current inventory situation and forecasts the demands for the next HHH periods. The decision-maker selects an order action with the objective of minimizing the total expected costs related to the inventory system.

The cost structure considered in the paper includes:

- ordering cost,
- inventory holding cost,
- shortage-related cost,
- future expected cost under uncertainty.

This formulation provides the foundation for the two main solution approaches used in the paper: the approximate dynamic programming benchmark and the reinforcement learning framework.

3.4 Benchmark Method: Approximate Dynamic Programming

Before the introduction of the DRL model, the paper has proposed a benchmark using the approximate DP method. This benchmark has significant importance because the quality of the learned DRL policy can be measured against this benchmark.

In the backorder problem, the proposed benchmark essentially follows the $(st, St)(s_t, S_t)(st, St)$ policy. The system places an order if the inventory position drops below a certain threshold. Once the order has been made, the inventory will be replenished up to the target level. This benchmark has been employed in the paper for the evaluation of the performance of the DRL-based policy in achieving near-optimal performance.

3.5 Reformulation as a Reinforcement Learning Problem

The main contribution of the paper is that it formulates the problem of inventory as a reinforcement learning environment. In a RL environment, an agent will interact with the environment. In each interaction, the agent will take a certain action and move to a new state.

The formulation of the RL environment is important because it allows the model to learn a replenishment policy through repeated experience instead of solving an optimization problem each time a decision is to be made.

3.6 State Representation

The state representation is certainly among the most significant aspects of the methodology. The state is intended to carry all necessary information to support inventory decisions when facing non-stationary demand. The state is comprised of:

- demand forecast information for the next HHH periods,
- current inventory level or inventory position,
- quantities already on order.

The inclusion of demand forecast directly within the state allows the agent to take into consideration future demand expectations, not just the current inventory level. This is certainly among the key aspects of the methodology that allows for non-stationary demand to be handled.

3.7 Action Representation

Rather than directly utilizing the raw order quantity as the action, the paper utilizes the (st,St)-based action representation. The actual replenishment quantity is determined by the environment based on the action selected by the agent.

The proposed action representation is particularly important because the optimal replenishment policy is discontinuous due to the fixed ordering cost. If the model were to directly predict the order quantity, the stability of the learning process would be threatened. The (st,St)-based action representation facilitates the learning process even when discontinuity exists.

3.8 Reward Function

The reward function in the RL environment is defined as the negative of the total cost incurred in each period. This includes:

- ordering cost,
- holding cost,
- shortage or lost-sales cost.

Since reinforcement learning aims to maximize cumulative reward, using negative cost as the reward means that the model effectively learns a policy that minimizes total long-run inventory cost.

3.9 Transition Dynamics

After the agent takes an action, the system moves to the next state according to the transition dynamics. This process includes:

- realization of demand,
- update of inventory level,
- shift of the forecast window,
- update of the order pipeline.

These transition rules ensure that the environment reflects the actual behavior of a rolling-horizon inventory system. As a result, the training process becomes more realistic and relevant to practical inventory management.

3.10 Randomized Training Environment

Another important aspect of the methodology is that it uses randomized initial states and also randomized demand forecasts. In other words, instead of training the agent using a specific type of demand, it is exposed to a number of different and randomized situations regarding its inventory and forecast.

The advantages of using this approach are twofold. It is advantageous because it allows for better exploration of the environment. It is also advantageous because it prevents overfitting to a specific type of demand. In other words, it ensures that the agent is able to generalize to unseen forecast scenarios.

3.11 PPO-Based Learning Algorithm

For training the DRL agent, the paper uses Proximal Policy Optimization (PPO). PPO is a widely used actor-critic reinforcement learning algorithm that is known for stable learning performance.

In this method:

- the actor learns the replenishment policy,
- the critic estimates the value of each state.

The use of PPO is appropriate in this study because it provides a stable way to learn continuous or parameterized actions in a complex inventory control environment.

3.12 Learning Challenge Caused by Fixed Ordering Cost

The paper also discusses an important methodological challenge related to **fixed ordering cost**. When an order is placed, the system experiences an immediate cost, but the benefit of that order may only appear after several periods when inventory arrives and future shortages are reduced. This delayed reward signal makes the learning problem more difficult.

To address this issue, the methodology uses:

- stronger critic training,
- warm-up strategies,
- smoother action representation through (s_t, S_t) .

These design choices help improve the stability of the learning process and make the DRL model more effective for inventory problems with discontinuous policy structures.

3.13 Extension to Forecast Evolution

The paper has extended the methodology beyond the rolling forecast horizon by the addition of the Martingale Model of Forecast Evolution (MMFE). Under the MMFE, the future demand forecasts can vary over a period of time based on the availability of information.

This is important because, in the real business scenario, the business does not operate on a single fixed forecast. Instead, the forecasts are constantly changing. The addition of the MMFE has extended the applicability of the proposed DRL framework to a dynamic business scenario where the forecast is changing over a period of time.

3.14 Computational Evaluation

After the DRL agent is trained, the learned policy is tested through computational experiments. The evaluation is conducted under three main settings:

- backorders,
- lost sales,
- MMFE-based forecast updating.

The policy obtained through the training of the DRL is then compared to the approximate dynamic programming benchmark. This comparison is used to ensure that the proposed approach is able to reach near-optimum performance and that it is able to perform well under harder inventory conditions.

3.15 Summary of the Methodological Process

Overall, the methodological process of the paper can be summarized in the following sequence:

1. define the non-stationary SISLSP,
2. formulate the inventory system mathematically,
3. construct the approximate DP benchmark,
4. reformulate the problem as a reinforcement learning environment,
5. define the state, action, reward, and transition structure,
6. train the DRL agent using PPO in randomized environments,
7. extend the framework to MMFE forecast evolution,
8. evaluate the learned policy against benchmark methods.

The fact that the methodology is explained through a series of steps renders it both systemic and useful. This methodology will allow the paper to test the viability of DRL as a potential tool for inventory optimization under varying demand conditions. For a better understanding, the entire methodological process of the study is represented through the following flowchart, where the major steps of the research are summarized: inventory formulation, benchmark development, reinforcement learning development, training, method extension, and evaluation.

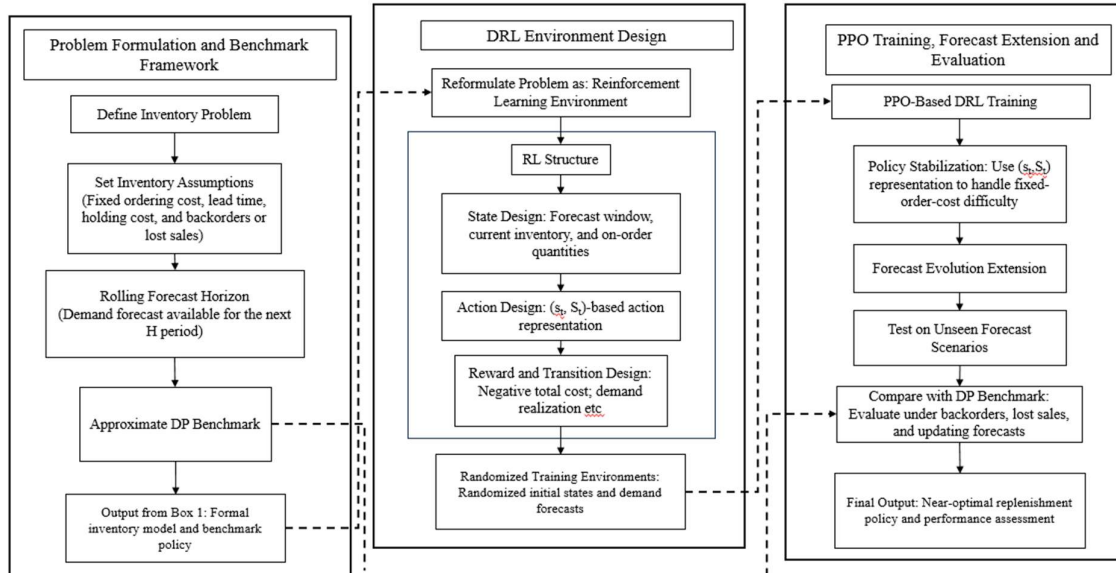


Figure 1: Overall research methodology of the paper, showing problem formulation, benchmark development, reinforcement learning environment design, PPO-based training, forecast extension, and performance evaluation.

4. Data Summary

The paper uses simulated data instead of real company data. The authors develop various types of inventory environments to test the efficacy of the deep reinforcement learning model in a non-stationary uncertain demand situation. The data used is mainly related to forecasts of demands, parameters of the inventory, and test scenarios.

For the main experiments, it is assumed that demand is normally distributed with a constant coefficient of variation over time. Training environments are created through variation of key parameters such as shortage backorder costs, coefficient of variation, fixed ordering costs, lead times, forecast horizons, and lost sales. The model is trained over 30 different parameter sets, with 10 agents per set. Initial inventories and expected demand means are generated randomly, and initial on-order quantities are set to zero.

For the test, the authors use 136-period demand forecasts and a variety of realistic trend patterns such as constant demand, linear growth, linear decline, and seasonal demand patterns. There is also additional experimental work on using the MMFE update procedure and a harder lost sales problem with Poisson demands. The data structure is such that it tests whether the model can perform well on a wide variety of different inventory situations and not just one specific situation.

Table 1. Detailed Summary of Data and Experimental Setup

Component	Details
Type of data	Simulation-generated experimental data
Problem setting	Single-Item Stochastic Lot-Sizing Problem (SISLSP) under non-stationary uncertain demand
Main purpose of data	To train and test the DRL model under different inventory and demand conditions
Main demand assumption	Gaussian demand distributions
Demand variability assumption	Same coefficient of variation (CoV) across periods

Component	Details
State-related data	Forecast means for upcoming periods, current inventory level, and on-order quantities
Training parameter variations	Backorder/shortage cost ((b)), coefficient of variation ((CoV)), fixed ordering cost ((K)), lead time ((L)), forecast horizon ((H)), and lost-sales condition
Training design	One parameter varied at a time, others kept at default values
Number of training settings	30 parameter combinations
Agents trained per setting	10 agents
Initial inventory	Randomly generated from a uniform distribution
Initial expected demand means	Randomly sampled from a uniform range
Initial on-order quantities	Set to zero
Test forecast length	136 periods
Test trend types	3 constant trends, 1 linear growth trend, 1 linear decline trend, and 3 seasonal trends
Purpose of test forecasts	To evaluate the trained model on unseen demand patterns
Additional forecast experiment	MMFE-based forecast updating
Additional lost-sales experiment	Poisson demand used to create more difficult shortage conditions
Benchmark comparison	DRL results compared with approximate dynamic programming policy

5. Answers to Research Questions

The paper demonstrates that deep reinforcement learning techniques can be successfully used to solve single-item inventory optimization problems with non-stationary uncertain demand. The authors claim that after appropriate training in randomly generated environments, the agents were able to generate near-optimal dynamic policies in real time for unseen rolling-horizon demand forecasts. This directly addresses the main research goal of the paper in a positive manner.

The study also validates the fact that the proposed DRL framework is capable of successfully incorporating forecast information. In the proposed DRL framework, future demand forecast information is incorporated into the state of the system, and the model is trained on a variety of randomized forecast environments. This enables the agent to learn how to respond to the changing demand patterns rather than assuming a stationary demand. The results show that the trained model is capable of interpolating good replenishment decisions even if the forecast pattern was not seen during the training process.

When comparing DRL with the benchmark policy, the paper finds that the results are generally very strong. In the backorders case, where the DP baseline is already a strong near-optimal method, the DRL policy still performs very closely to the benchmark and provides competitive results. This means the proposed method is able to match high-quality traditional approaches while also offering the advantage of fast real-time decision making after training.

In the lost-sales case, the findings are even more interesting. The paper reports that the DRL-based policy can significantly outperform the approximate DP baseline when lead times are sufficiently long. This is an important result because lost-sales problems are harder than backorder problems, and it suggests that DRL may be especially useful in more complex inventory settings where traditional heuristics become weaker.

The paper also answers the question of whether the method can work under forecast updating dynamics. The authors extend the methodology to the Martingale Model of Forecast Evolution (MMFE) and show that the DRL policy can still be a preferable alternative to the DP baseline, even when forecast variability is high. This indicates that the proposed approach is not limited to simple fixed-forecast settings.

Overall, the answers to research questions are positive. The paper concludes that the proposed PPO-based DRL method is able to learn good replenishment strategies for non-stationary

inventory systems, performing almost optimally or even better than traditional benchmark methods in many situations. Thus, it is possible to state that the research is supporting the application of DRL methods to real-world inventory management tasks.

6. Future Research Directions

The paper also opens several avenues for future research. First, it is possible to expand the suggested methodology to more complex forecast dynamics than those used in the paper. Although it is already demonstrated how to apply MMFE-based forecast updating in the paper, it is possible to consider even more complex and realistic patterns of demand evolution.

Second, it is suggested that DRL should be applied to more complex inventory control problems. The paper demonstrates that the approach is already promising in lost sales scenarios, particularly when lead times are high. It is possible to explore more complex scenarios in which traditional dynamic programming heuristics are not very effective.

Third, future studies may attempt to apply the method to larger, more practical supply chain systems, such as multi-item or multi-echelon inventory systems. Since the paper is specifically on SISLSP, the extension of the method to larger supply chain systems would make it more useful. This is also in line with the overall DRL literature, as presented in the paper.

Finally, future studies may attempt to improve the learning method by creating a more stable learning algorithm for inventory systems with fixed ordering costs and discontinuous policy. The paper has already established the learning difficulty of the problem; hence, it is an important area for future research.

In other words, the paper suggests that the future research should be focused on more general forecast models, more difficult inventory problems, larger supply chain systems, and more effective DRL training approaches. These areas can make the suggested methodology more useful in real environments.

7. Implementation in the Context of Bangladesh

7.1 Introduction

The methodology developed in the chosen paper also has good potential for application in Bangladesh, as in most supply chains, demand is unstable, conditions for replenishments are uncertain, and decision support is lacking. The paper under review discusses a single-item inventory system with non-stationary uncertain demand, rolling forecasts, a fixed ordering cost, a known lead time, backorders, or lost sales. The conditions considered in the paper are very common in most industries in Bangladesh, as demand is rarely stable in such industries, and decisions to order goods are taken in a hurry in most cases.

However, it is not possible to use this model directly without modifications. The main reason is that the paper is designed and tested within a simulated environment. On the contrary, the environment for Bangladeshi firms is full of noise, fragmented information, infrequent forecast updates, logistics issues, and lack of analytics support. Thus, the main task here is not the direct use of the model but rather adaptation. The main task for the Bangladeshi context is to design the DRL model framework by incorporating demand drivers, lead time behavior, cost factors, data preparation, and deployment.

In this context, the Bangladesh implementation should be treated as a decision-support system development project rather than a purely academic model application. The aim should be to preserve the strength of the original framework while making the inputs, assumptions, and deployment process suitable for the operational realities of Bangladesh.

7.2 Relevance of the Model for Bangladesh

The selected model is relevant to Bangladesh because inventory-related uncertainty is a common feature across local supply chains. Demand often changes because of festivals, seasonal shifts, promotions, inflationary pressure, weather conditions, transport disruption, or sudden changes in consumer purchasing behaviour. At the same time, many firms still depend on manual reorder rules, experience-based judgment, or simple spreadsheet-based planning. Such methods may work under stable conditions, but they become weak when demand and lead time change simultaneously.

The relevance of the paper lies in three major aspects. First, the model is forecast-aware. Since the state representation includes demand forecasts over a rolling horizon, it is more realistic than methods that rely only on current inventory position. Second, the model is adaptive. The paper shows that a trained DRL policy can respond to changing demand patterns instead of assuming stationarity. Third, the model is computationally attractive after training, because ordering decisions can be generated in real time without repeatedly solving an optimization model. This feature is valuable in Bangladeshi settings where quick operational decisions are needed but advanced optimization capability is limited.

Therefore, the model is especially relevant in Bangladesh for firms that face repeated replenishment decisions, demand volatility, service-level pressure, and uncertain supply conditions.

7.3 Sectors Where the Model Can Be Applied

7.3.1 FMCG Sector

FMCG industry is considered to be one of the strongest industries for implementing the suggested model in the context of Bangladesh. The demand for FMCG products fluctuates significantly during Ramadan, Eid, promotional activities, weather conditions, and geographical demand variations. The industry should focus on ensuring the availability of products in the market while avoiding stockholding in different stock points. DRL can be considered as a solution for better ordering decisions for high-volume products with fluctuating demand pattern over time.

7.3.2 Pharmaceutical Supply Chain

The pharmaceutical industry is another industry where stock-out and stock-in issues are significant because they involve costs. In the case of stock-out, it can influence the treatment and service levels, while in the case of stock-in, it can result in stock becoming obsolete, lock up working capital, and inefficiency in storage space. In Bangladesh, supply chain issues in the pharmaceutical industry can also be impacted by transportation disruption, increased demand during outbreaks, and differing replenishment lead times. A forecast-aware inventory model can be beneficial in this industry, especially in the case of essential drugs and frequently demanded products.

7.3.3 Retail and E-Commerce

Retail businesses or e-commerce businesses usually experience a surge in demand during promotional campaigns, discount offers, or during festive seasons. Demand can vary based on locations or customer segments. The rolling horizon feature of the proposed model is applicable to such businesses.

7.3.4 Imported Industrial Materials and Spare Parts

Products imported may face uncertain lead times due to custom delay, shipping delay, and transport difficulties. In such a scenario, the methodology proposed in the paper is more relevant because it has already taken into consideration the lead times and the decision-making process.

7.4 Key Challenges in the Bangladesh Context

Although the methodology is promising, implementation in Bangladesh would face several practical obstacles.

7.4.1 Poor Data Quality

One major concern in Bangladesh is the quality and reliability of data used for operation. In this paper, it is based on simulation data, which means that all variables, cost, and demand are consistent with each other. However, it is possible that many Bangladeshi companies may experience incomplete sales records, delayed inventory updates, incomplete data on suppliers, and many other problems with their data, which might lead to an unstable decision-making pattern learned by the DRL policy.

7.4.2 High Demand Volatility

Demand fluctuations in Bangladesh are not only seasonal but also event-driven. Product sales might fluctuate due to events such as Ramadan and Eid, Pohela Boishakh, school opening season, political unrest, monsoon conditions, heat waves, and geographical demand patterns. Therefore, it is possible that a model trained on generic trend variation might not learn the true causes of demand fluctuations. This means that the Bangladesh variant of the model should be extended with explicit demand signals.

7.4.3 Uncertain Lead Time

Lead time in Bangladesh is often unstable because of traffic congestion, transport bottlenecks, supplier inconsistency, customs delay, port congestion, weather-related interruptions, and occasional labour or political disruptions. In many industries, average lead time alone is not enough to describe the replenishment process. The model must therefore consider variability, delay probability, and the operational consequences of late arrivals.

7.4.4 Limited Technology Readiness

Another challenge is that not all companies are yet ready to take advantage of advanced DRL systems. This is particularly true for small and medium-sized companies that may not yet have an ERP system, digital inventory records, and analytics capabilities in house. In larger companies, there may also be an organizational separation between forecasting and replenishment processes. Therefore, it is important to follow an incremental approach from a pilot environment with limited scope.

7.5 Modifications Needed for Bangladesh

To make the methodology practical for Bangladesh, several modifications should be introduced.

7.5.1 Local Demand Features Should Be Added

The initial framework should be enhanced to include local explanatory variables in the demand input. In the context of Bangladesh, demand can be driven by festivals, promotional activities, religious seasons, school opening periods, monsoon conditions, temperature changes, income variations, and social upsets. The variables should be represented as additional features to ensure that the model does not solely depend on demand history and forecast values.

7.5.2 Lead Time Should Be Modeled as Variable

In the original paper, the lead time is incorporated while setting the inventory. However, while adapting the model for Bangladesh, it should be considered a variable/stochastic component. The adapted model should account for the average lead time, lead time dispersion, risk of severe delay, supplier reliability, and risk of transport disruption. This will make the replenishment policy more realistic.

7.5.3 A Data Cleaning Stage Should Be Added

Before model training begins, firms should establish a formal data-preparation stage. This stage should include removal of duplicate records, correction of obvious data entry errors, treatment of missing values, alignment of transaction dates, reconciliation of stock movement with warehouse balances, and aggregation of demand to a consistent planning interval such as daily or weekly buckets. Event-based and seasonal indicators should also be created at this stage. Without such preprocessing, the DRL model may capture noise instead of actual operational patterns.

7.5.4 Implementation Should Start with High-Priority Items

The paper discusses a single item stochastic lot sizing problem. The appropriate starting point for a small group of high-priority items in a practical implementation in Bangladesh is rather a subset of SKUs rather than all SKUs at once. The starting point could be A-class inventory items, essential medicines, high stockout cost items, or imported items with fluctuating supply, for instance.

7.5.5 Cost Structure Should Be Expanded

The basic structure of the cost includes ordering cost, holding cost, and shortage or lost sales cost. In the case of Bangladesh, other components of the cost structure should be included so that the learned policy is more realistic in its implications to the business process. These other components include emergency transport cost, spoilage or expiry risk, service level penalty, customer dissatisfaction, sudden increase in procurement price, and delayed fulfillment cost. An extension to the reward structure will be more representative in real business scenarios.

7.5.6 Human Oversight and Override Logic Should Be Added

For the Bangladesh system, it is not recommended to operate as a completely autonomous decision maker. Instead, a recommendation from DRL would be provided to the planners with a system in which overriding is possible. The managers would have the capability to change decisions related to order recommendations in situations where extraordinary events happen, such as a strike, flood conditions, import restrictions, or a supplier shutdown. This would help in gaining trust in the system and make it more acceptable.

7.6 Summary Table of Required Modifications

Table 2. Summary Table of Required Modifications

Area	Original Paper	Bangladesh Adaptation
Demand input	Non-stationary rolling forecast	Add festival, weather, promotion, and regional demand features
Lead time	Included as a model parameter	Treat as variable or stochastic with delay risk
Data source	Simulation-generated environment	Use cleaned business data from sales, inventory, and suppliers
Product scope	Single-item problem	Start with high-priority or ABC-classified items
Cost structure	Ordering, holding, shortage/lost-sales cost	Add emergency transport, spoilage, and service-level related cost
Forecast revision	Rolling forecast and MMFE	Add local event-based forecast updates
Deployment style	Experimental evaluation	Pilot in one warehouse, one branch, or one product group
Decision governance	Model-generated policy	Add planner review and override mechanism

7.7 Proposed Implementation Framework

A practical implementation framework in Bangladesh should move in stages rather than all at once.

Stage 1: Problem Scoping and Product Selection

The firm should first define the business problem clearly. This includes selecting the product category, warehouse or distribution region, service-level target, and current performance baseline. High-priority products should be chosen for pilot deployment.

Stage 2: Data Collection and Preparation

Data such as historical sales, inventory levels, pipeline orders, supplier lead times, stock-out occurrences, and relevant external demand drivers should be recorded. This data should then be preprocessed, standardized, and transformed for training the models.

Stage 3: Forecast and Feature Engineering

The rolling horizon demand forecasts should be generated based on business data that is available. At this stage, Bangladesh-specific characteristics such as Ramadan effects, promotional periods, regional effects, monsoon effects, and disruption markers should be included in the state representation.

Stage 4: Model Training and Policy Testing

The DRL agent would then be trained with localized inputs and realistic cost assumptions. The trained policy would be tested offline before live deployment against the current reorder rules used by the firm, as well as a benchmark heuristic similar in nature to what has been discussed in this paper.

Stage 5: Pilot Deployment

The system should initially be deployed in a restricted environment, such as a single warehouse, product family, branch network, or region. During this period, the DRL recommendation should be compared with the existing planning method, and human review should remain active.

Stage 6: Evaluation and Controlled Scale-Up

If the pilot results indicate an improvement, the firm can scale the system to more products and locations. Scaling the system should be based on the actual benefits realized in terms of stockout reduction, service level, inventory efficiency, and replenishment stability

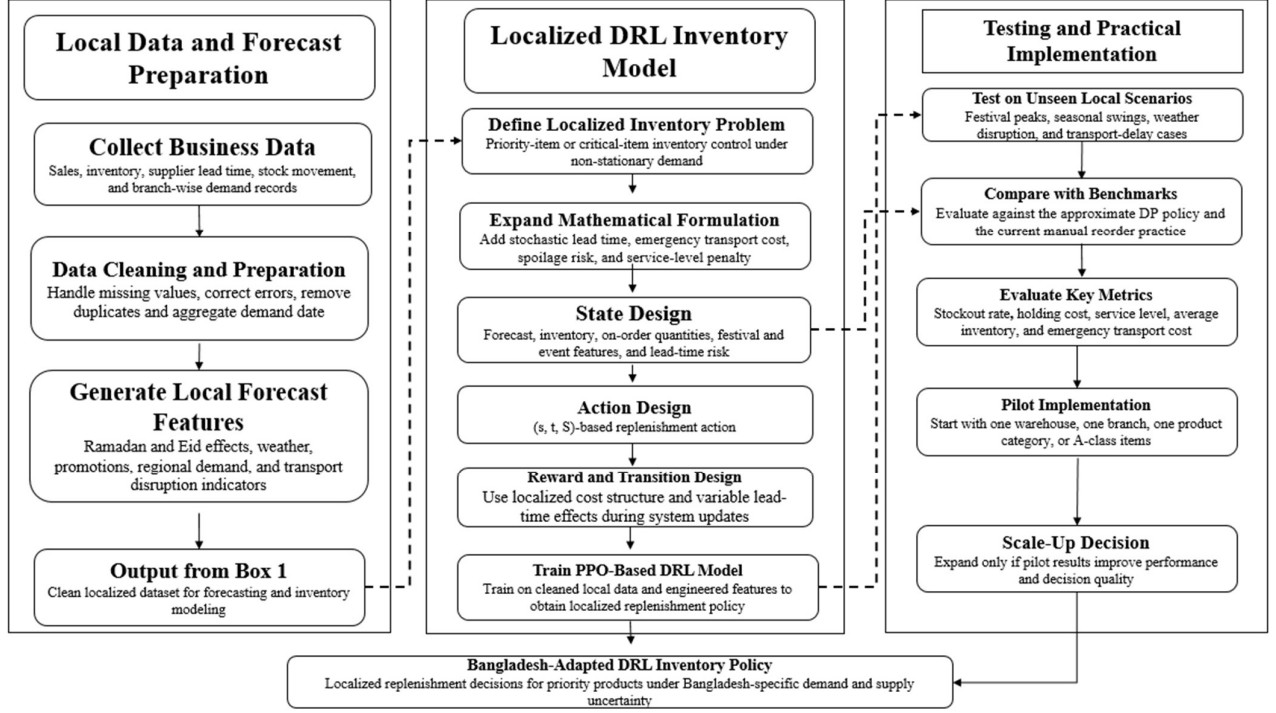


Figure 2: Bangladesh-adapted implementation framework of the proposed DRL-based inventory model for inventory optimization under local demand and supply uncertainty.

7.8 Monitoring and Performance Evaluation

Bangladesh implementation should not be considered a success simply because mathematically reasonable order quantities are generated by the model. Operational performance measures should be considered for judging the success of the system in a Bangladesh implementation. Some of the performance measures are the stock-out frequency, fill rate or service level, average inventory levels, inventory turns, emergency replenishments, holding costs, shortage costs, and total replenishment costs. Additionally, the quality of forecast revision and lead-time deviation should be tracked, as the model relies on good input information in a timely manner.

Periodic system retraining is also recommended for the system in a Bangladesh implementation. This is because the demand conditions in Bangladesh may fluctuate as a result of inflation, changes in government policies, new entrants in the industry, changes in logistics, and changes in customer behavior.

7.9 Expected Benefits and Practical Risks

If properly adapted, the proposed DRL framework could generate several benefits in Bangladesh. It could improve responsiveness to changing demand, reduce stockout occurrences, lower excessive inventory, support more consistent replenishment decisions, and reduce dependence on purely manual judgment. It may also be particularly useful in environments where the same ordering decision must be made repeatedly under changing forecasts.

However, practical risks remain. Poor data quality can weaken the learned policy. Limited trust from managers may delay adoption. Unexpected disruptions can make historical patterns less informative. There is also a risk that firms may expect immediate full-scale automation without first building the necessary data and governance structure. Therefore, implementation success will depend not only on the algorithm but also on disciplined data management, pilot-based deployment, and organizational acceptance.

7.10 Overall Assessment

Thus, in conclusion, the selected paper is appropriate in conceptualizing Bangladesh because it considers non-stationary demands, rolling horizon forecasts, lead time, and computationally efficient replenishment decisions via deep reinforcement learning. These factors are similar to the challenges being faced in supply chain management in Bangladesh. However, the model should not be implemented in its original form without any modifications because the selected paper is based on the structured simulation environment, which cannot be implemented in the imperfect and disrupting real-world scenario in Bangladesh.

Thus, the best approach would be to keep the basic structure of the DRL model proposed in the selected paper and modify its data structure, cost structure, lead time, governance structure, and stages of implementation to suit the Bangladeshi scenario. Once the model is modified according to the Bangladeshi scenario, it can be implemented in the Bangladeshi supply chain to help Bangladeshi firms in decision-making regarding inventory management under uncertain scenarios. In this way, the selected paper is not only academically relevant but also practically implementable in Bangladesh.

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